

Nama: Nashira Salima Firmansyah
NIM: J0403251056
Kelas: TPL A2

```
print("Nama : Nashira Salima Firmansyah")
print("NIM : J0403251056")
print("="*40)

class Node:
    def __init__(self, data):
        self.left = None
        self.right = None
        self.data = data

def insert(root, data):
    if root is None:
        return Node(data)

    if data < root.data:
        root.left = insert(root.left, data)
    else:
        root.right = insert(root.right, data)

    return root

def inorderTraversal(root, result):
    if root:
        inorderTraversal(root.left, result)
        result.append(root.data)
        inorderTraversal(root.right, result)

def preorderTraversal(root, result):
    if root:
        result.append(root.data)
        preorderTraversal(root.left, result)
        preorderTraversal(root.right, result)

def postorderTraversal(root, result):
    if root:
        postorderTraversal(root.left, result)
        postorderTraversal(root.right, result)
        result.append(root.data)

# =====
# DATA BERDASARKAN NIM
# =====
```

```

root_value = 56
data = [36, 76, 26, 46, 66, 86, 41]

root = Node(root_value)
for d in data:
    root = insert(root, d)

# =====
# HASIL TRAVERSAL
# =====

inorder_result = []
preorder_result = []
postorder_result = []

inorderTraversal(root, inorder_result)
preorderTraversal(root, preorder_result)
postorderTraversal(root, postorder_result)

# Output
print("Data Tree :", [root_value] + data)
print()

print("In-order   :", inorder_result)
print("Pre-order   :", preorder_result)
print("Post-order  :", postorder_result)

```

Screenshot Output

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\nashi\OneDrive\Desktop\IPB\SEM 2\ASD_J0403251056_Nashira-Salima-Firmansyah> & C:/Users/nashi/python.exe "
Nama : Nashira Salima Firmansyah
NIM  : J0403251056
=====
Data Tree : [56, 36, 76, 26, 46, 66, 86, 41]

In-order   : [26, 36, 41, 46, 56, 66, 76, 86]
Pre-order  : [56, 36, 26, 46, 41, 76, 66, 86]
Post-order : [26, 41, 46, 36, 66, 86, 76, 56]
PS C:\Users\nashi\OneDrive\Desktop\IPB\SEM 2\ASD_J0403251056_Nashira-Salima-Firmansyah> |

```

Hasil traversal:

- In-order : [26, 36, 41, 46, 56, 66, 76, 86]
- Pre-order : [56, 36, 26, 46, 41, 76, 66, 86]
- Post-order : [26, 41, 46, 36, 66, 86, 76, 56]

Penjelasan singkat:

- Apa perbedaan ketiga traversal tersebut?
 - In-order: hasil urut, cara kerjanya adalah: kiri → root → kanan
 - Pre-order: lihat struktur awal, cara kerjanya adalah: root → kiri → kanan
 - Post-order: Biasa digunakan untuk menghapus tree. cara kerjanya adalah: kiri → kanan → root
- Menurut anda, Traversal mana yang paling mudah digunakan untuk melihat struktur awal tree?
 - Pre-Order. Root / data pertama akan muncul di awal, dibandingkan kedua traversal lainnya
- Apakah hasil traversal akan sama jika data dimasukkan dengan urutan berbeda?
 - Tidak, hasil traversal tidak akan sama jika urutan data berbeda. Hal ini karena bentuk Binary Search Tree bergantung pada urutan data yang dimasukkan. Perubahan struktur tree akan mempengaruhi hasil traversal.